

Lecture 01: Course Introduction

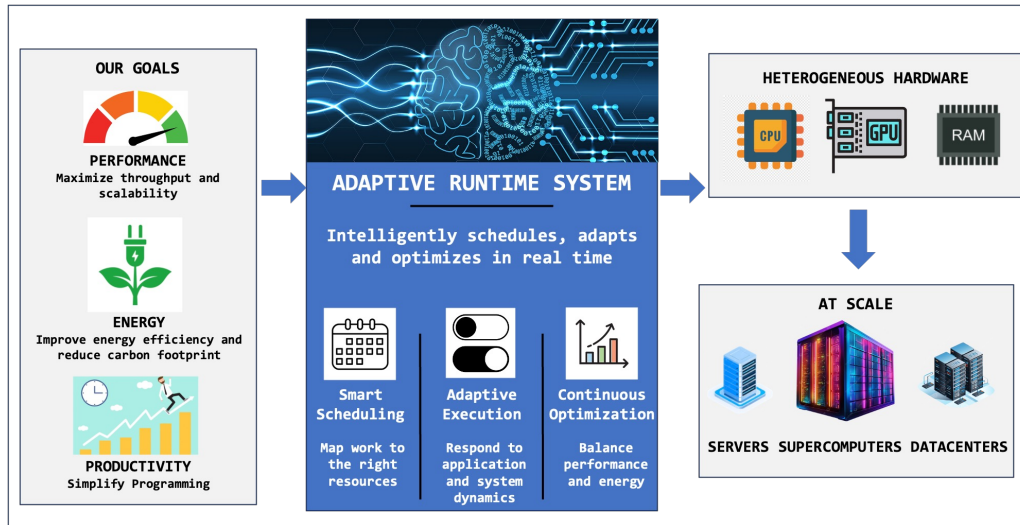
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About me



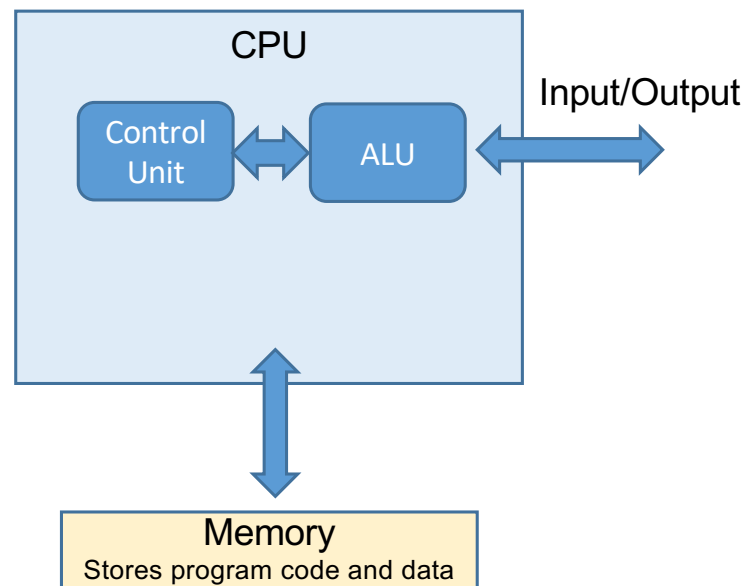
I spend my time either at the HiPeC Lab, or trekking in the Himalayas..

Today's Lecture

- Modern computing architecture
- High-level overview of operating systems
- Roles of an OS
- Challenges in modern OS
- Course evaluation and logistics

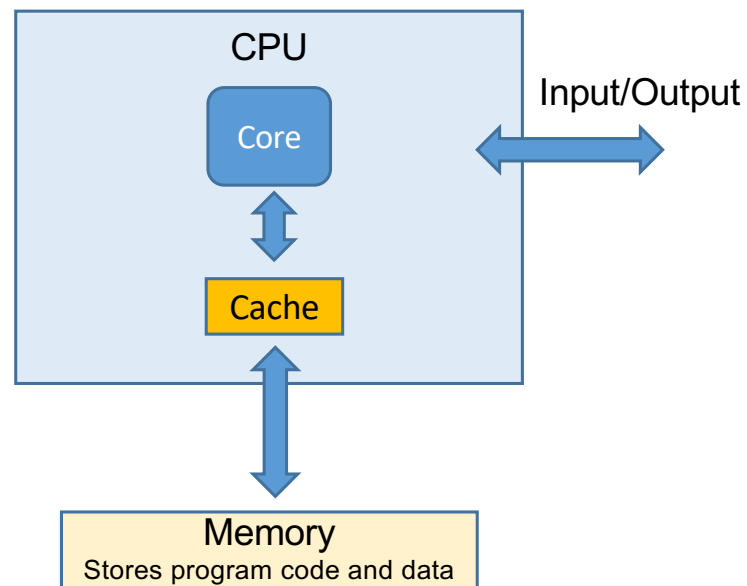
**Let us first quickly try to understand
the modern computer architecture**

Von Neumann Architecture & Associated Issues (1/4)



- John Von Neuman in 1945 came up with the architecture for computers that we even use today (albeit with several changes)
 - Prior to that, Harvard architecture used to exist
 - Separate storage for instructions and data
- **Next Problem?**
 - Memory bottleneck
 - Access latency to memory quite high
 - High CPU stalls while fetching code and data

Von Neumann Architecture & Associated Issues (2/4)



- **Solution**

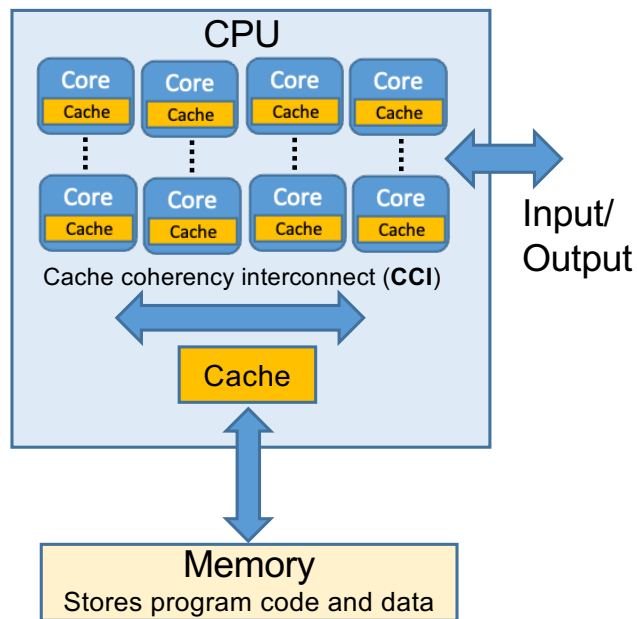
- Add a small cache memory inside the processor to store frequently accessed memory

- **Next Problem?**

- Performance bottleneck

- Design issues with increasing the performance of single core processor
 - High heat dissipation
 - Even capable of melting the processor!
 - High power consumption

Von Neumann Architecture & Associated Issues (3/4)



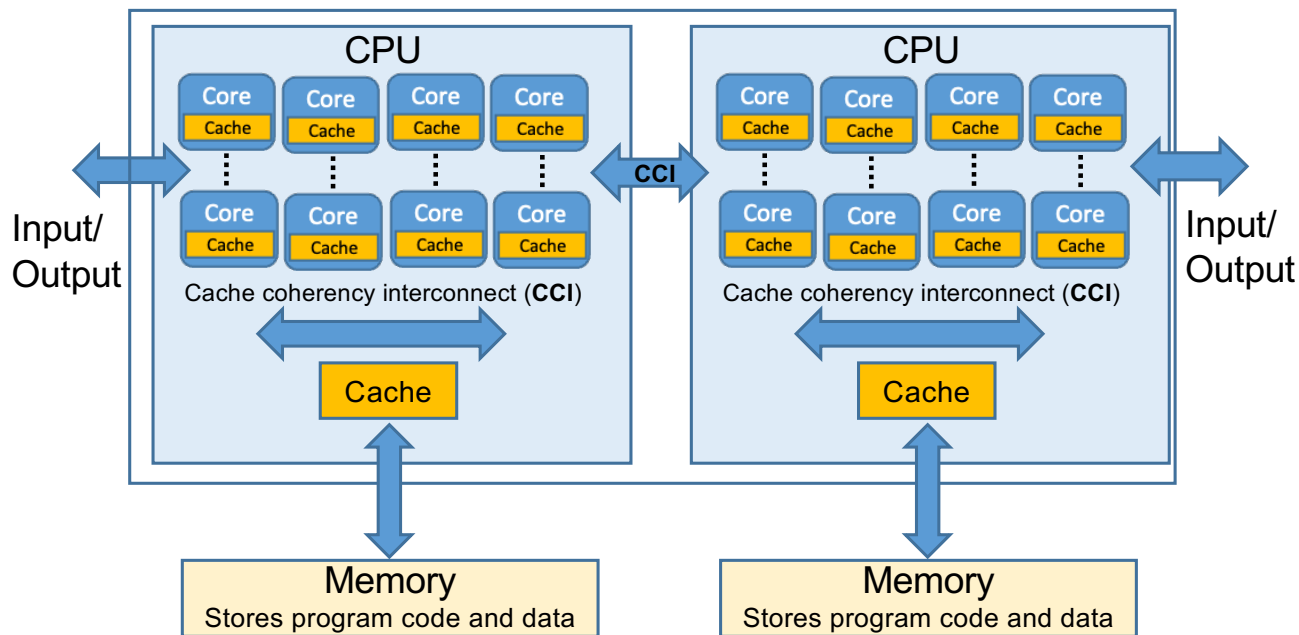
● Solution (around 2004)

- Add more cores to achieve better performance instead of increasing the performance of a single core
 - Still maintains the Moore's law
- Add cache coherency interconnect (CCI) to fetch data on one core from the other core's cache instead of going all the way up to main memory

● Next problem

- Performance bottleneck
- Modern applications requires more **performance** and **data**
 - How to improve the performance and data storage of a single system?

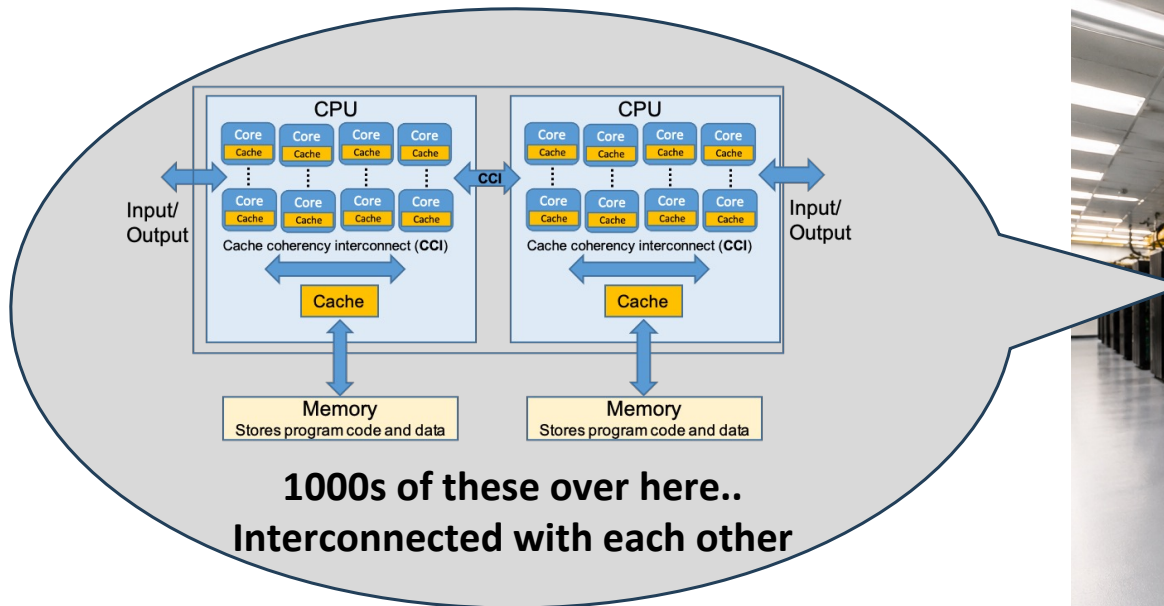
Von Neumann Architecture & Associated Issues (4/4)



● Solution

- Add more processors (a.k.a. sockets) on the same motherboard
- Provide local memory banks at each socket
 - Each CPU can access local and remote memory banks

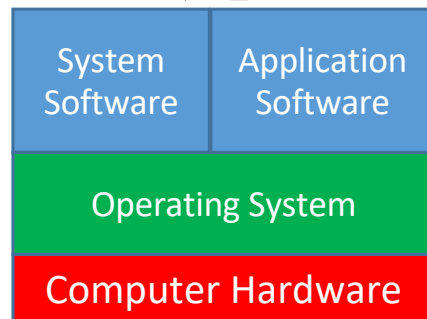
Supercomputers (More Computation + More Data)



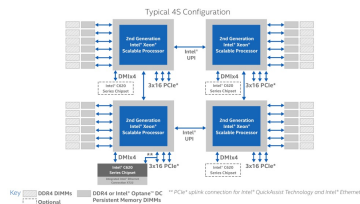
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What is an Operating System?



- OS is a piece of software whose job is to manage the computer's resources for its users and applications



Analogy: The Hardware



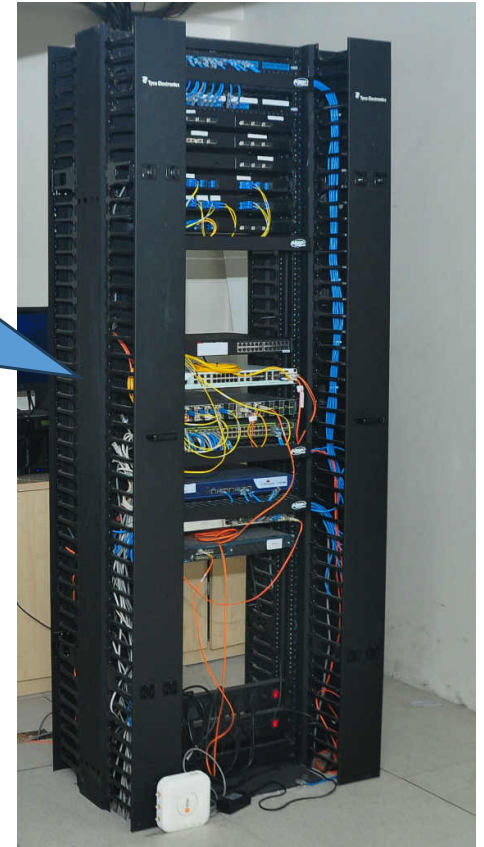
Analogy: Disk & DRAM

Library – persistent storage of data (Disk)



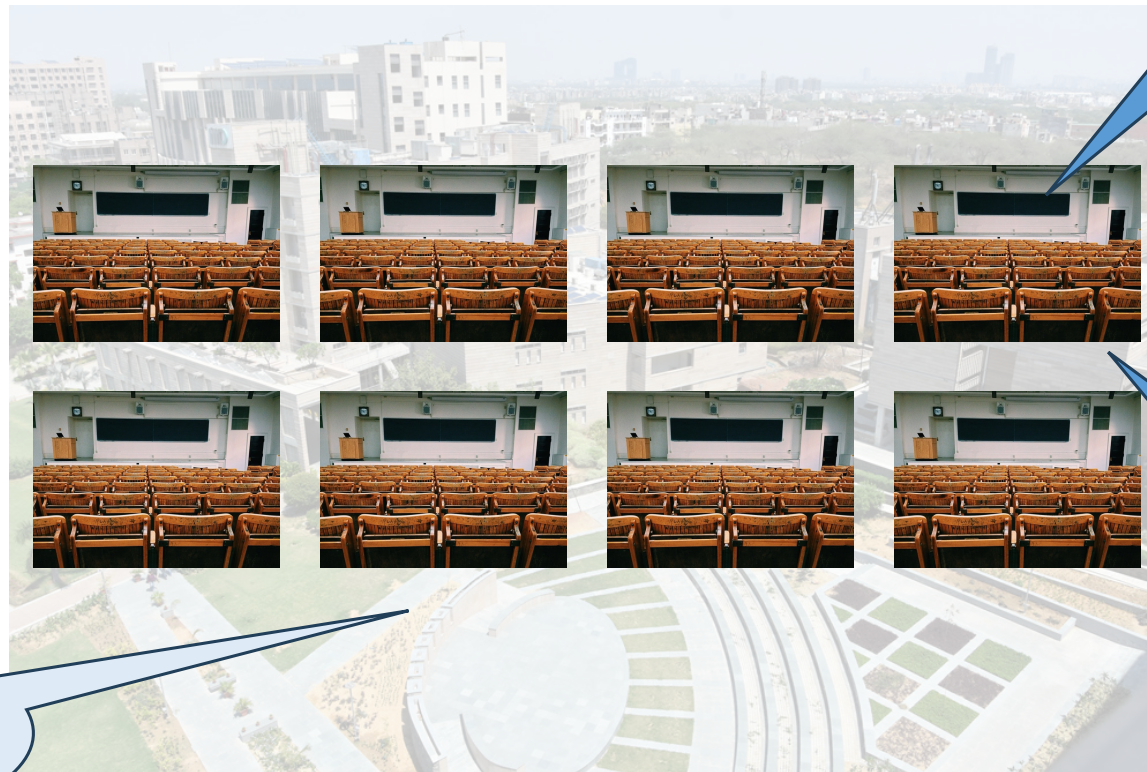
Data is read from the Disk (**Library**) and stored in central server (**DRAM**) in the form of **lecture slides**

Any familiarity with memory hierarchy?



Analogy: The Multicore CPU

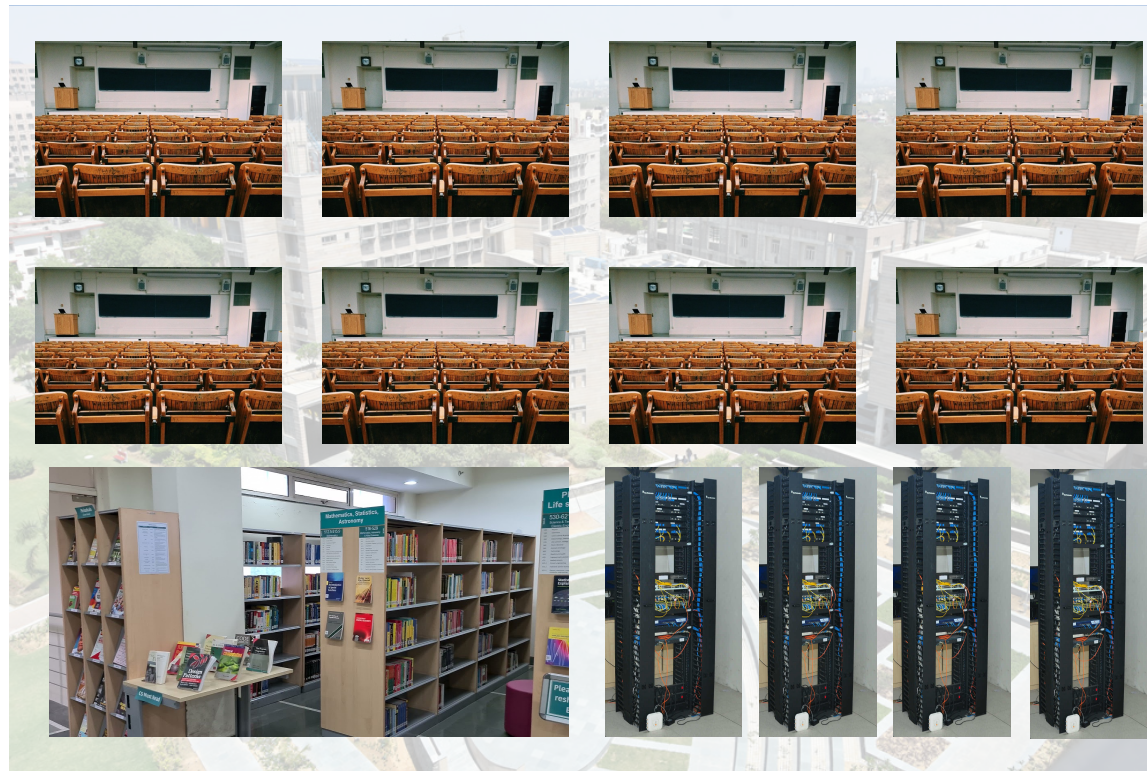
Board (Cache)



Fixed number
of Cores

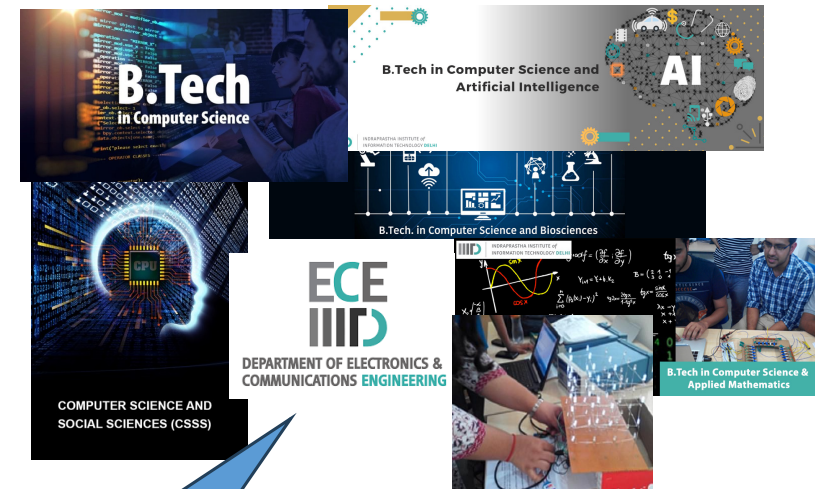
Classroom
(Core)

Analogy: The Complete Hardware



Analogy: Operating System & Applications

OS: Faculties
and Staff



Applications:
Several
BTech/MTech
programs

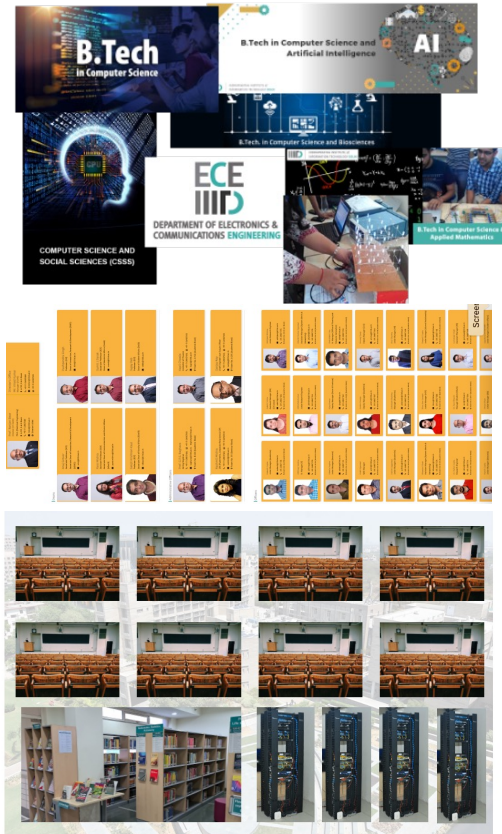
Each **application**
has several **Tasks** to
complete (courses)

Major Goals of Operating Systems

1. Virtualization
 - CPU
 - Memory
2. Concurrency
3. Persistence
4. Protection and security

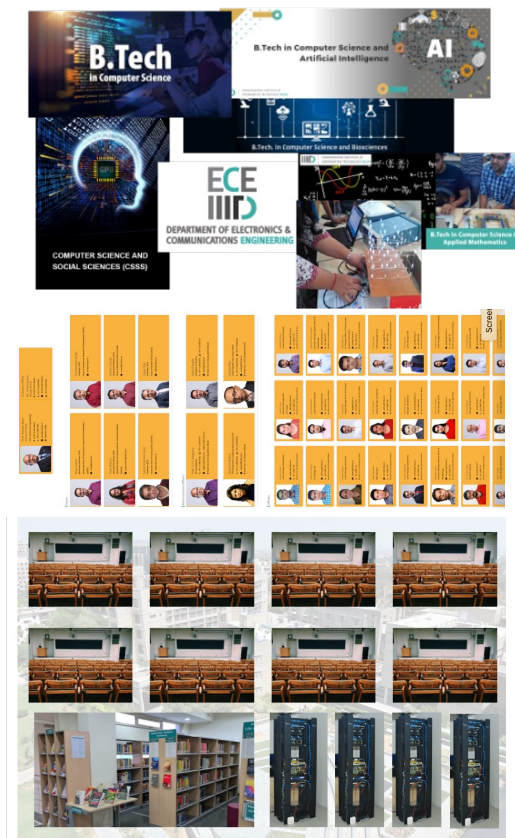
Analogy: Virtualization (CPU cores)

- The operating system gives an “**illusion**” to each application that the entire hardware is only for that application
 - So many applications can execute on the same hardware with limited resources (fixed number of cores, DRAM, and disk)



Scheduling:
Time table is prepared for each semester for smooth running of each program

Analogy: Virtualization (CPU cores)

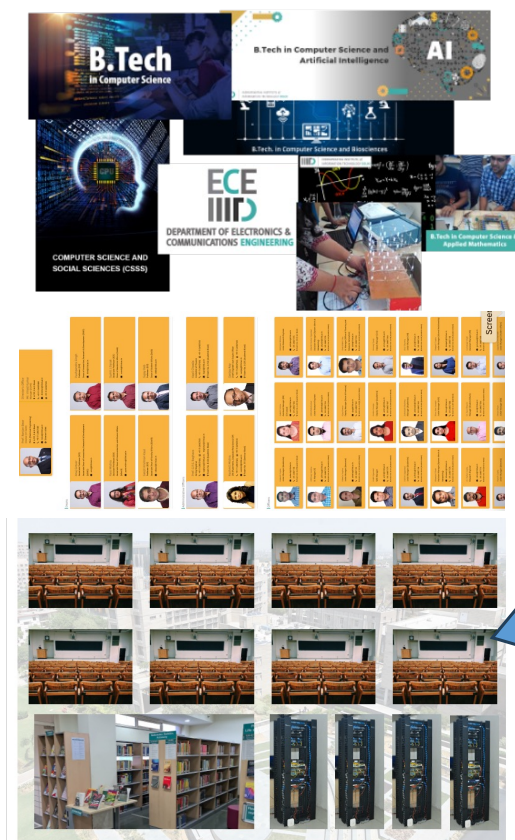


Every
program has
many courses,
each with
several
lectures

- Application stored in file v/s running applications
- Each running application (**Process**) has some code and data associated with it (e.g., course and lecture materials)
 - A faculty member prepares a lecture by fetching textbooks from the library (disk) and then preparing lecture slides, which are stored on a server (DRAM)
 - Lectures are **Scheduled** in a classroom where the faculty uses the whiteboard (**Cache**) to teach the topic

Analogy: Virtualization (CPU cores)

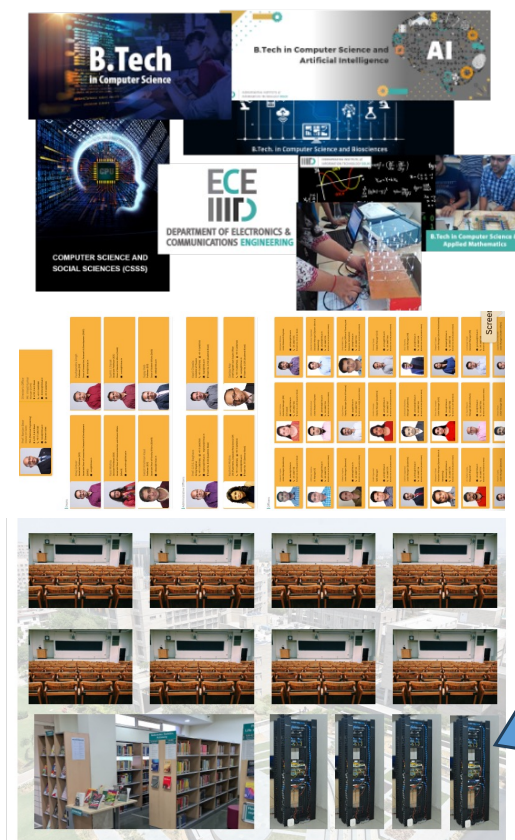
- Only one **Task** can run on a CPU core at any given time
 1. A classroom is used to run lectures from different courses
 2. During the lecture, students write down notes from the board (**cache**) onto their notebook (persistent storage)
 3. Next faculty enters the classroom for delivering a different lecture
 4. First, a recap of the topics discussed in the last lecture is carried out before starting the lecture
 - Reloading the context
- Above steps are a very high-level overview of **context switch between processes for time sharing of a CPU core!**



Several lectures are taught using a fixed number of classrooms

Analogy: Virtualization (Memory)

- Faculties and admin (OS) stores the data from each program (courses, lectures, documents, etc.) in the servers (DRAMs)
- It gives an illusion that there are dedicated memory spaces for each program
 - In the reality each program's data could be stored in pieces over several DRAMs (physical address)
 - However, this complexity is abstracted away by the O.S.



There are several servers (DRAMs) to store the data associated with each program

Analogy: Concurrency

Scheduling algorithm
of IIT-Delhi

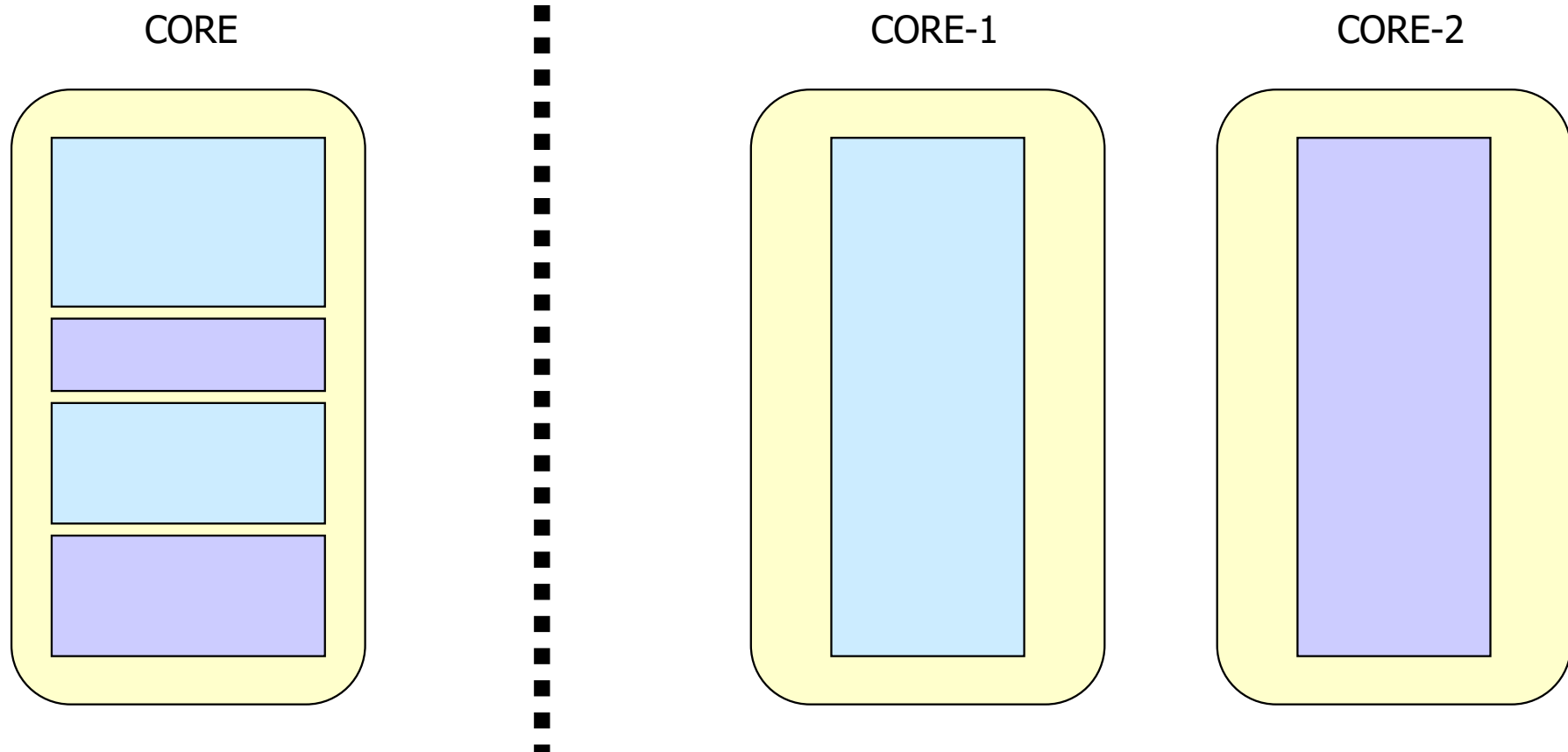
B.Tech. 3rd & 4th Year, M.Tech. 1st & 2nd year, Ph.D. - Monsoon Semester AY 2022-23														
Day	8.30-9.00	9-9:30	9:30-10	10-10:30	10:30-11	11-11:30	11:30-12	12-12:30	12:30-1	1-1:30	1:30-2	2-2:30	2:30-3	3-3:30
Mon			CN (Sec A)	C21	Slot 4	AC	C03	Slot 2						DIP
			CN (Sec B)	C102		DMG	C21							
			BML	C03		QMD	C216							
			WSI	C13		ETB	C22							
			IDUDA	C208		ME	A007							
			NDM	C22		IAGA	C208							
			MLBA	A006		DFML	C209							
			CMSMR	C209		BioP	C214							
			MAD	C214		ADL	A006							
			RA-II	C215										
Tue			CoMeG	C03	Slot 1	FOE	C02	Slot 5						
			ML (A)	C11		CGAS	A006							
			ML (B)	C21		IIA	C213							
			LST	C13		SDOS	A106							
			QM	C22		DSc	C210							
			TML	C208		CMOS	C211							
						SC	A007							
			AERM	C215		FF	C21							
			DAVP	C24		ADC	C12							
			WN	C213		WARDI	C208							
			NDMA	C210		G&M	C209							
			CN (Sec A)	C21	Slot 4	CA	A006	Slot 9						DIP
			CN (Sec B)	C102		Chi	C13							

B.Tech. CSE program has several courses, some of which could be scheduled together in the same slot, but in different classrooms

B.Tech. CSE program
has several courses,
some of which could be
scheduled together in
the same slot, but in
different **classrooms**!

- A program could have several independent tasks (set of instructions) that could be executed simultaneously over multicore CPUs
 - Multiple threads of execution
- However, some course has prerequisite courses that the students should complete first
 - Dependencies in multi-threaded execution

Concurrency vs. Parallelism



Analogy: Persistence

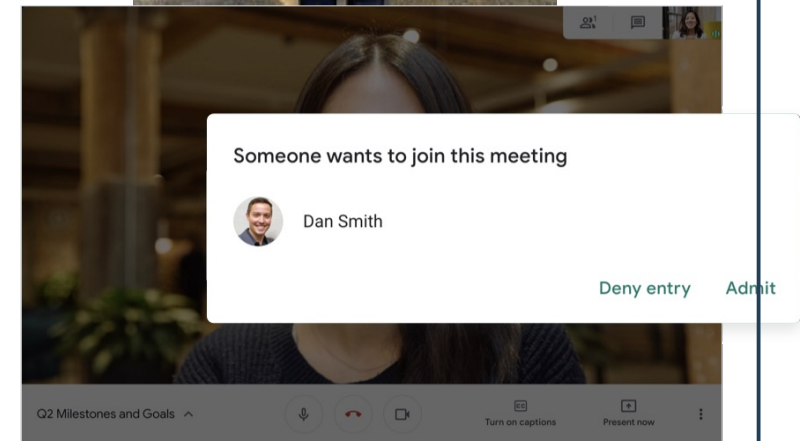
- At the end of each semester, academic department collect and permanently save the lecture slides, quiz sheets, assignments and project, and student's answer sheets for mid/end semester exams (for each courses)

Analogy: Protection (Internal) & Security (External)



Course Name	Course Acronym	Course Code
<u>Computer Architecture</u>	CA	CSE511/ECE511

Each program can only access its own data (courses) unless there is some authorization obtained by the O.S



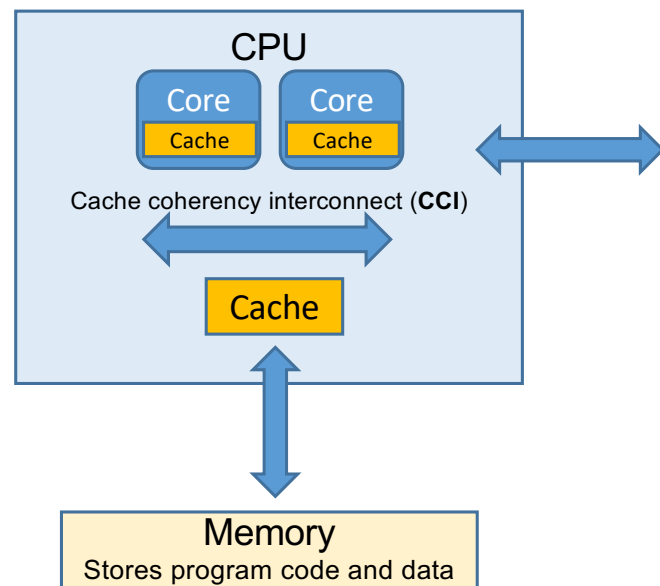
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Roles of an Operating System

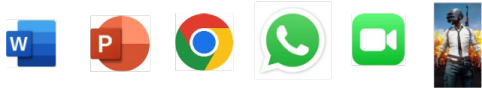
- Referee
 - Manages all the shared resources in the computer
- Illusionist
 - Each application thinks that the entire computer belong to itself
- Glue
 - Offers standard services to simplify application development and facilitate sharing

Roles of an Operating System



- Let us try to understand it from the perspective of a desktop/laptop

Roles of an Operating System



● Referee

- Each application will be launching several tasks
 - Gaming app will require graphic rendering, network play support, taking player input via keyboard/mouse, etc.
 - Zoom will require tasks such as network connection, access to mike/camera, graphic rendering, etc.
- As there are several cores, each task can run on a different core
 - But, there are only finite number of resources (cores, caches, input/output devices, memory etc.)



● Illusionist

- Allows the application to have hundreds of tasks (several times more than available cores), infinite amount of memory (DRAMs are only in few GBs), etc.

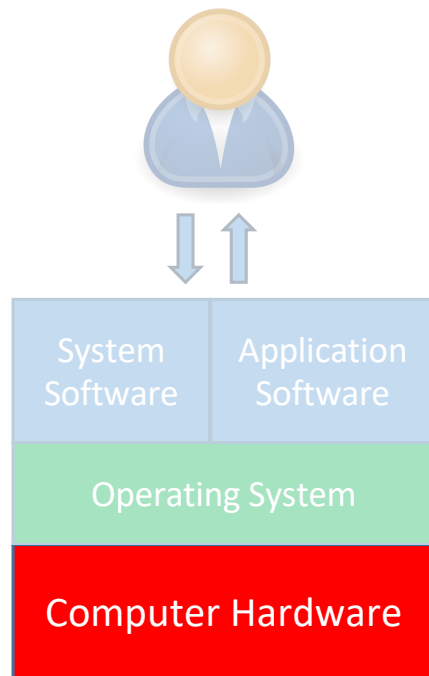
● Glue

- Standard APIs for network connection, accessing input/output devices, etc.

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Operating System Specific to Hardware

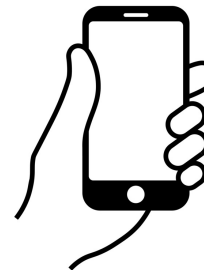


- Throughput oriented (manages several tasks)
- E.g., Windows, Linux, MacOS
- Lightweight
- Highly user friendly (GUI)
- Smaller memory footprint (more than RTOS)
- E.g., Android, iOS
- Time bound and limited tasks
- May work with small memory
- E.g., Nucleus RTOS

Laptop/Desktop
Operating System



Mobile Operating
System



Real time Operating
System

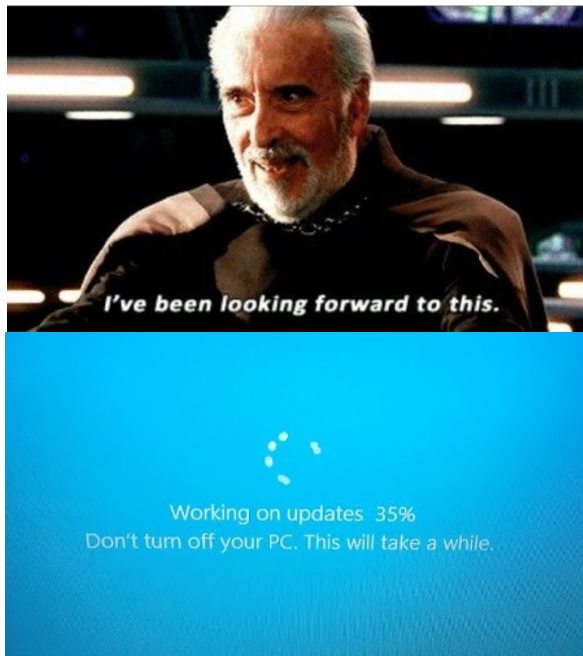


Major Challenges in Modern OS

- Dependability

Me: *restarts computer because it's lagging*

Windows Update:



Predictable? Reliable?
(frequent bug fixes, updates...)

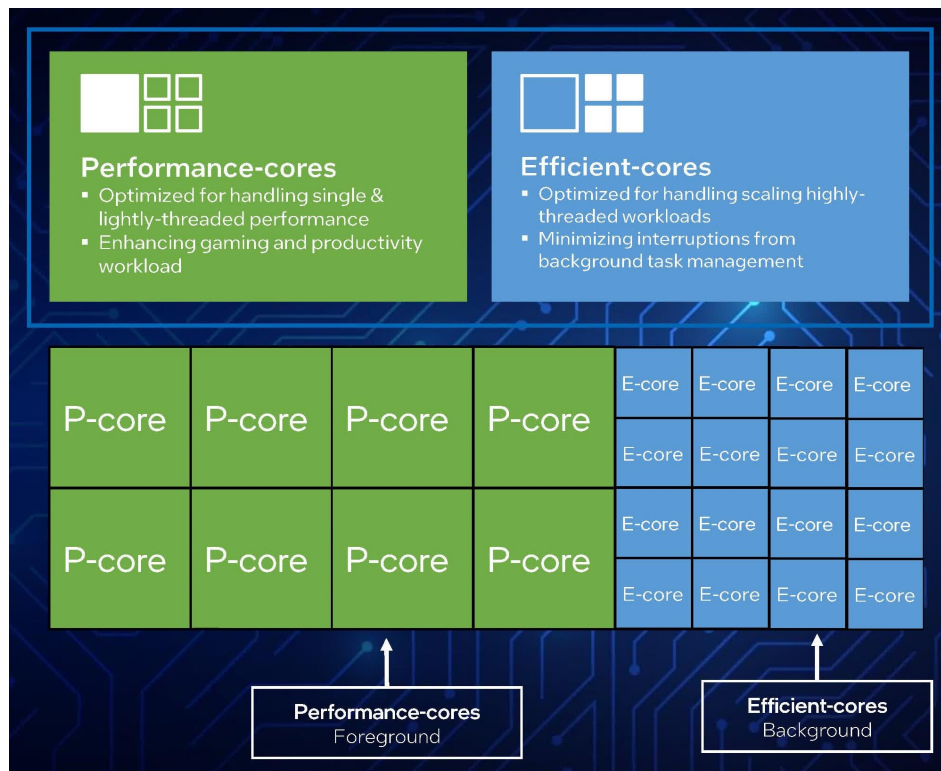
Predictable?
Reliable?

V/S



Major Challenges in Modern OS

● Performance and Energy



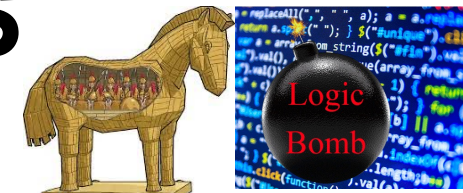
Which tasks should run on P-core and which ones on E-cores?

What should be the processor core frequency while running a particular task? High frequency can give better performance, but requires more energy, and vice-versa

Images source: <https://www.techspot.com/news/94919-preview-core-i9-13900-engineering-sample-performance-looks.html>

Major Challenges in Modern OS

- Protection and Security



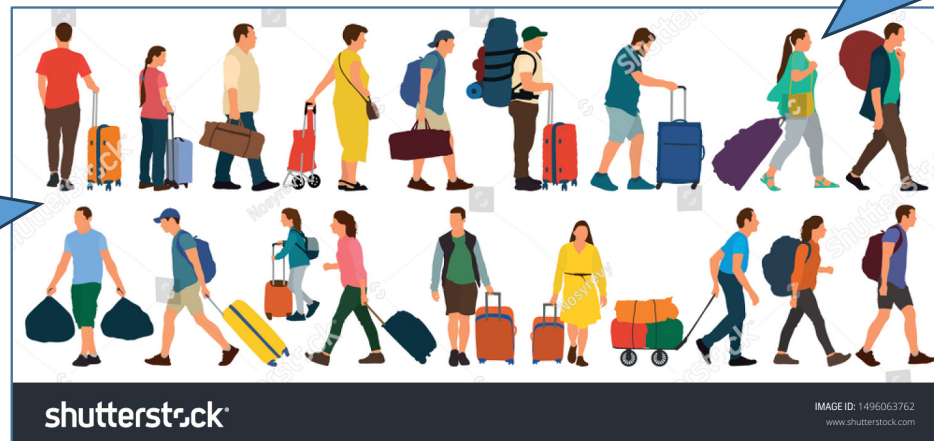
They are
lacking
Security



Let's unleash
our arsenal

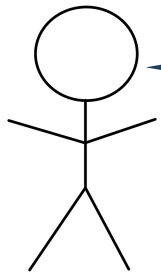
We are
Protected

Each application can only access
its own **baggage** (files, memory,
CPU). They should not access
another application's **baggage**
unless they have proper
authorization



Major Challenges in Modern OS

● Portability



I am quite **Portable**. I just need to change my shoes



I am quite **Portable** too!
I just need to change my machine interfacing code in kernel



Summary

- Every design decision you saw today is a trade-off
 - Dependability vs. frequent updates/bug fixes
 - Performance vs. energy consumption
 - Protection/Security vs. overhead of enforcing it
 - Portability vs. squeezing out hardware-specific performance
- One single implementation of OS can't be optimize for all four challenges at once
 - Every OS you use (Linux, Android, RTOS) has picked a different balance

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Course Evaluation (Section-A)

- **Both sections will be graded/evaluated SEPARATELY**
 - Below mentioned evaluation policy is specific to Section-A only
- Quiz: 10% (N-1 policy)
 - No surprise quizzes
 - Will be held during lecture hours (around 20mins duration)
- Semester exams
 - Mid-sem: 20%
 - End-sem: 35%
- Take home assignments 35%
 - **NO (N-1 policy) !!!**

I might have a different lecture plan/content than you see currently on Techtree. However, the COs would remain the same

Important Information

1. We will **Not** upload mid/end semester solution/rubric on Google Classroom
 - Although, we will discuss it in class
2. We will **Not** upload assignment solution/rubric on Google Classroom
 - Although, TAs will provide relevant information during the demo
3. We will be **Not** be taking in-class attendance
 - Lecture recordings will not be provided
4. There will be some surprise in-class bonus quizzes

Course Prerequisites

- **C programming and debugging skills are must!**
 - You must have used C in OS refresher module
 - If you don't know C then better start practicing so as you don't face issues with the course assignments
 - First one will be released early next week!

We will strictly follow the IITD plagiarism policy. No excuses if you get caught in plagiarism

You must declare explicitly in your assignments if your submission includes any AI-generated/internet code

Course Logistics

7kqikpww

- Only **Section-A** students should register to above classroom
 - Section B students would be removed from GC of Section A
- Machines for assignments
 - You can use your laptop
 - Would require Linux OS (or any VM running Linux OS)
 - In case you have problems please feel free to first shoot an email to TF (Ayush), and then to me if your issues isn't resolved
- Course textbook is Operating Systems Three Easy Pieces (OSTEP)

Next Lecture

- Source code to machine code